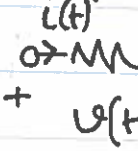
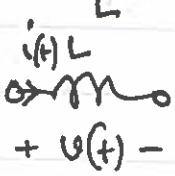
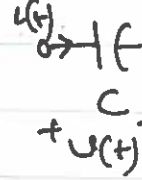
Lecture 2

Suggested Reading : 2.1 - 2.3 (Pai).
Today : * Single-phase ac circuits.
* Concepts of power in ac circuits.
* Phasors.

A. Let us start modeling a real-world example using passive, linear circuit elements.

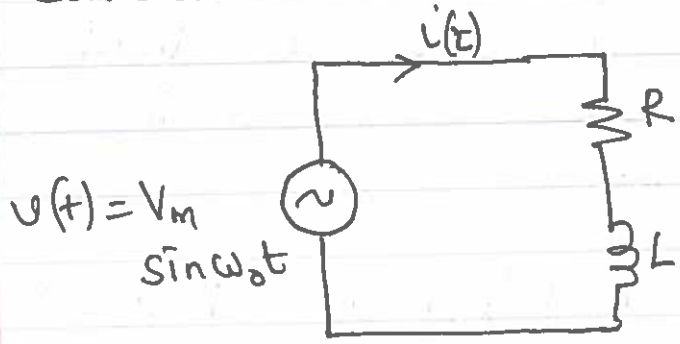
What are our options ?

Fourier ↑ Transform.

Elements	Time-domain	Frequency domain
1. Resistance 	$v(t) = i(t) \cdot R$	$V(j\omega) = RI(j\omega)$
2. Inductance 	$v(t) = L\frac{di(t)}{dt}$	$V(j\omega) = j\omega L\, I(j\omega)$
3. Capacitance 	$i(t) = C\frac{dv(t)}{dt}$	$V(j\omega) = \frac{1}{j\omega C} I(j\omega)$

where $V(j\omega) = F[v(t)]$
$I(j\omega) = F[i(t)]$.

Example: We want to model a toaster connected to an ac wall plug:



Two fundamental questions — (a) Can we compute the current?
(b) Can we " the power drawn by the toaster?

Two ways to answer:

A. Time domain:

$$R\,i(t) + L\frac{di(t)}{dt} = v(t) = V_m \sin \omega_0 t \quad \text{—(1)}$$

$\Rightarrow$ valid for all time.

Assuming we are in sinusoidal steadystate, ** which implied we neglect all switch transients,

i.e $v(t+T) = v(t)$
$i(t+T) = i(t)$

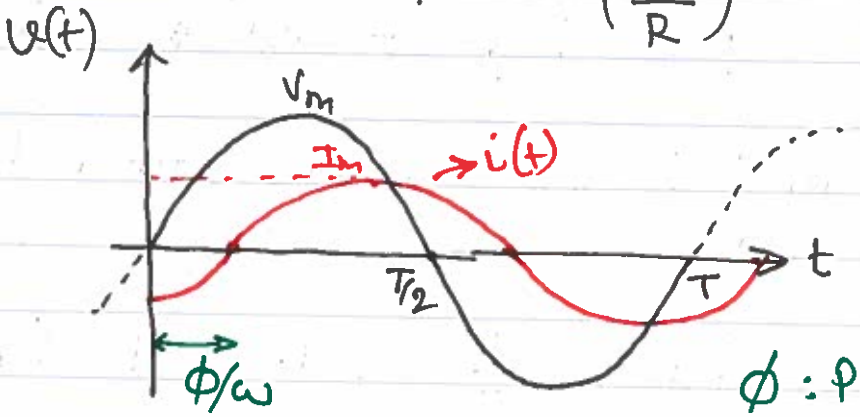
where $T = \frac{2\pi}{\omega_0}$; $\omega_0 = 2\pi \times 60$ rad/s → in U.S
$= 2\pi \times 50$ rad/s → in Europe.

$V_m = 120\sqrt{2}$ V
in U.S
$= 230\sqrt{2}$ V
in Europe.

let us call this as I_m.

$$i(t) = \frac{V_m}{\sqrt{R^2+(\omega_0 L)^2}} \sin(\omega_0 t - \phi)$$

where $\phi = \tan^{-1}\left(\frac{\omega_0 L}{R}\right)$



ϕ : Phase angle.

In this example, current is lagging voltage.

Q. How about power?

Instantaneous power, $p(t) = v(t) \cdot i(t)$

or, $p(t) = V_m \sin\omega_0 t \cdot I_m \sin(\omega_0 t - \phi)$

$$= \frac{V_m I_m}{2}\left[\cos\phi - \cos(2\omega_0 t - \phi)\right]$$

$$= \frac{V_m I_m}{2}\left[\cos\phi - \cos 2\omega_0 t \cos\phi - \sin 2\omega_0 t \sin\phi\right]$$

$$= \frac{V_m I_m}{2}\left[\cos\phi\right]\left\{1 - \cos 2\omega_0 t\right\} - \frac{V_m I_m}{2}\sin\phi \sin 2\omega_0 t$$

For a moment let us call:

$$P = \frac{V_m I_m}{2} \cos\phi \; ; \; Q = \frac{V_m I_m}{2} \sin\phi .$$

$$\therefore \quad p(t) = P(1 - \cos 2\omega_0 t) \mp Q \sin(2\omega_0 t) .$$

So, what is the average power drawn by the load ?

$$\langle p(t) \rangle = \frac{1}{T} \int_0^T p(t)\, dt = P . \triangleq \text{Active power.}$$

This power is utilized by the toaster and is converted into heat
– can be utilized. (active power)

Case I:
Suppose $R = 0 ; \tan\phi = \frac{\omega_0 L}{R} = \infty \quad \therefore \phi = 90°$ (lag).

$$\therefore \quad p(t) = - Q \sin(2\omega_0 t)$$

average is zero, the instantaneous power goes back and forth between source and load.

Q. $\triangleq$ Reactive power
– does not do useful work.

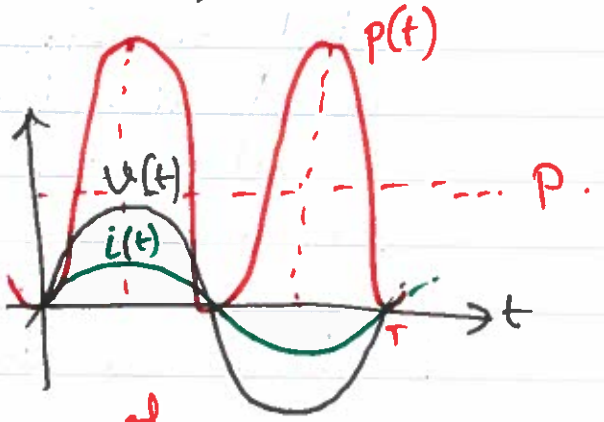
Note: $P^2 + Q^2 = \left(\frac{V_m I_m}{2}\right)^2 \triangleq S$ (apparent power).

Case II:

Suppose, $L = 0$; $\tan\phi = 0 \therefore \phi = 0°$.
voltage and current are in phase.

instantaneous, $p(t) = P(1 - \cos 2\omega_0 t)$

average, $\langle p(t) \rangle = P$



Frequency of instantaneous power: twice the fundamental frequency.

Bottomline: $p(t) = v(t) \cdot i(t)$ ← instantaneous power

$P = \frac{V_m I_m}{2} \cos\phi$: Active/average power/real power.

$Q = \frac{V_m I_m}{2} \sin\phi$: Reactive power.

Sometime its convenient to express P and Q in terms of rms qty.

$V_{rms} = \frac{V_m}{\sqrt{2}}$; $I_{rms} = \frac{I_m}{\sqrt{2}}$

Try it at home: $x_{rms} = \sqrt{\frac{1}{T}\int x^2 dt}$

$x = v(t)$ or $i(t)$.

$\cos\phi$ is defined as the power factor

It expresses how well we are utilizing the wiring infrastructure. For example, the wiring is rated for V_{rms}, fuse is rated for I_{rms}. To utilize the wiring, we would like draw as much active/average (useful) power as possible and minimize sloshing of power back and forth.

B. Frequency domain:

Fourier transform of (1). (complex number)

$$R\,I(j\omega) + j\omega L\,I(j\omega) = V(j\omega).$$

$$I(j\omega) = \frac{V(j\omega)}{R + j\omega L}$$

So, we can represent is polar form,

$$V(j\omega) = V_m \angle 0^\circ = \text{phase}$$

(V_m: magnitude; 0° implies reference.)

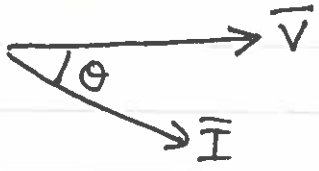
Recall $v(t) = V_m \sin\omega_0 t$

$$I(j\omega) = I_m \angle \theta \leftarrow \text{some angle}.$$

(I_m: magnitude)

This leads to our phasor notation,

Voltage phasor : $\bar{V} = \left(\frac{V_m}{\sqrt{2}}\right)\angle 0^\circ$



$\bar{I} = \left(\frac{I_m}{\sqrt{2}}\right)\angle \theta^\circ$

↖ magnitude of the phasor is usually expressed as rms.

$$\bar{I} = \frac{V_{rms}\angle 0^\circ}{R + j\omega L} = \frac{V_{rms}}{\sqrt{R^2 + \omega^2 L^2}}\angle \theta$$

where $\theta = -\tan^{-1}\left(\frac{\omega L}{R}\right)$ ← lagging power factor